

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.3 TAXONOMY II: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How can we identify and classify living organisms accurately? You will write a Claim-Evidence-Reasoning (CER) Final Explanation using what you observed in Lessons 1 to 8 about the organisms around our school: maize, bean plant, grass, fern, moss, mushroom, mosquito, butterfly, earthworm, snail, chicken, lizard and goat. Answer each section in the space below it. Use evidence from your own observations, your group work and your Driving Question Board. Use your own words and record scientific terms in English. You may discuss ideas in Kiswahili or your mother tongue before you write. Then use your answers to help your group build its Local Biodiversity Classification Guide.

Part 1: The problem with our own groupings

In Lesson 1 your group sorted the organisms from around the school and compared your grouping with other groups in a gallery walk. (a) What rules did different groups use, and what did you notice when you compared the groupings? (b) Explain why a standard system of classification is needed so that any person anywhere can identify and study an organism.

Write your answer here:

Part 2: Classifying plants

Use the plants you examined (moss, fern, maize and bean plants). (a) Name the three plant groups you studied and state the observable features that place a plant in each group. (b) Explain why grouping plants by these features is more reliable than grouping them by size, colour or use as food.

Write your answer here:

Part 3: Classifying animals

Think about the animals in our collection (earthworm, snail, mosquito, butterfly, chicken, lizard, goat). (a) Choose three animals from different phyla and write a CER statement placing each in its phylum, using observable body features. (b) Explain how the goat, chicken and lizard are further separated within their phylum.

Write your answer here:

Part 4: Dichotomous keys

(a) Describe how a dichotomous key is built and used. (b) Write the first two couplets of a key you could use to separate three organisms from our collection. (c) Explain why different groups using the same key reach the same answer, unlike your Lesson 1 groupings.

Write your answer here:

Part 5: Viruses and the final explanation

(a) What are the characteristics of viruses, and give two examples of why they are important in agriculture, health or biotechnology. (b) Now write your Final Explanation for the driving question 'How can we identify and classify living organisms accurately?' Use a claim, evidence from at least two lessons, and reasoning. Say how your thinking has changed since your Lesson 1 model.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim answers the driving question	Gives a clear claim that a standard system based on shared, observable characteristics and dichotomous keys allows accurate identification and classification.	Gives a claim about classification but it is general or only partly linked to observable characteristics and keys.	Claim is missing, unclear or does not answer the driving question.
Evidence from the lessons	Uses specific evidence from several lessons, such as the school organisms, moss, fern, maize, bean, earthworm, snail, mosquito, goat, chicken and lizard, and the key results.	Uses some correct evidence from the lessons but it is limited or lacks detail.	Evidence is vague, incorrect or not taken from the lessons.
Scientific reasoning and use of terms	Explains why observable shared features give the same result for everyone while size, colour or use do not, and uses divisions, phyla, classes, couplets and	Gives some reasoning and uses most terms correctly.	Little or no reasoning; terms are missing or misused.

	viruses correctly.		
Dichotomous key and virus knowledge	Writes correct contrasting couplets and accurately describes virus characteristics and their importance with Kenyan examples.	Writes a mostly workable key and gives some correct virus facts.	Key is not workable and virus ideas are missing or incorrect.
Revision of thinking	Clearly compares the Lesson 1 groupings with the final model and states a relevant new question.	Mentions that thinking changed but with limited comparison.	Does not show how thinking has changed.

ATTRIBUTION AND LICENSING

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